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UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

Master's Thesis

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Thesis as a requirement to obtain the degree of
Magister en Ciencias de la Ingeniería Electrónica

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Abstract

The global maritime industry is increasingly prioritizing sustainability, promoting the adoption of various hybrid and fully electric propulsion systems for different vessel types. This paper presents a design and sizing methodology of an energy storage system for a hybrid tugboat, customized to meet the operational demands while minimizing environmental impact. The proposed framework integrates analysis of load profile, energy storage sizing, battery technologies selection, and propulsion system optimization to achieve an optimal balance between performance, and emissions reduction. The results demonstrate that a well-designed energy storage system for a hybrid tugboat can achieve reductions in fuel consumption and greenhouse gas emissions without compromising the availability and robustness required for towing and transit operations.

Resumen

La industria marítima mundial está dando cada vez más prioridad a la sostenibilidad, promoviendo la adopción de diversos sistemas de propulsión híbridos y totalmente eléctricos para diferentes tipos de buques. En este artículo se presenta una metodología de diseño y dimensionamiento de un sistema de almacenamiento de energía para un remolcador híbrido, personalizado para satisfacer las exigencias operativas y minimizar el impacto medioambiental. El marco propuesto integra el análisis del perfil de carga, el dimensionamiento del almacenamiento de energía, la selección de tecnologías de baterías y la optimización del sistema de propulsión para lograr un equilibrio óptimo entre el rendimiento y la reducción de emisiones. Los resultados demuestran que un sistema de almacenamiento de energía bien diseñado para un remolcador híbrido puede lograr reducciones en el consumo de combustible y las emisiones de gases de efecto invernadero sin comprometer la disponibilidad y la robustez necesarias para las operaciones de remolque y tránsito.

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Chapter 1

Introduction

1.1 Motivation and Background

The growing global concern over climate change has significantly pressured industries to adopt sustainable practices. The maritime sector, responsible for approximately 2.5% of energy sector CO_2 emmisions as shown in figure 1.1, has continuosly increased in emissions [1] [2].

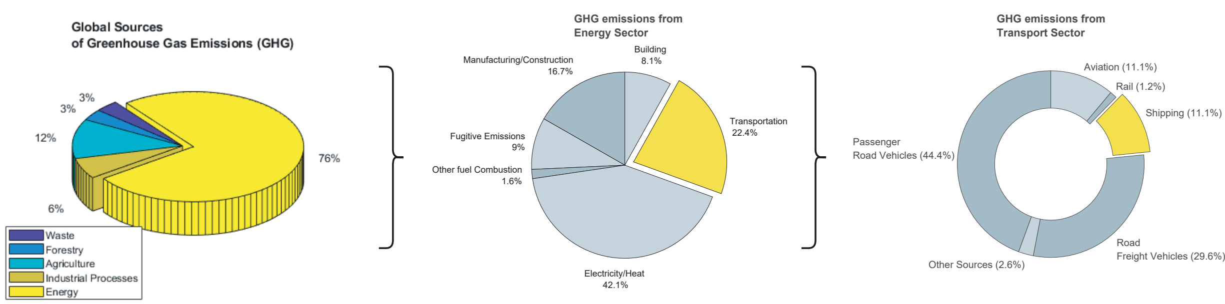
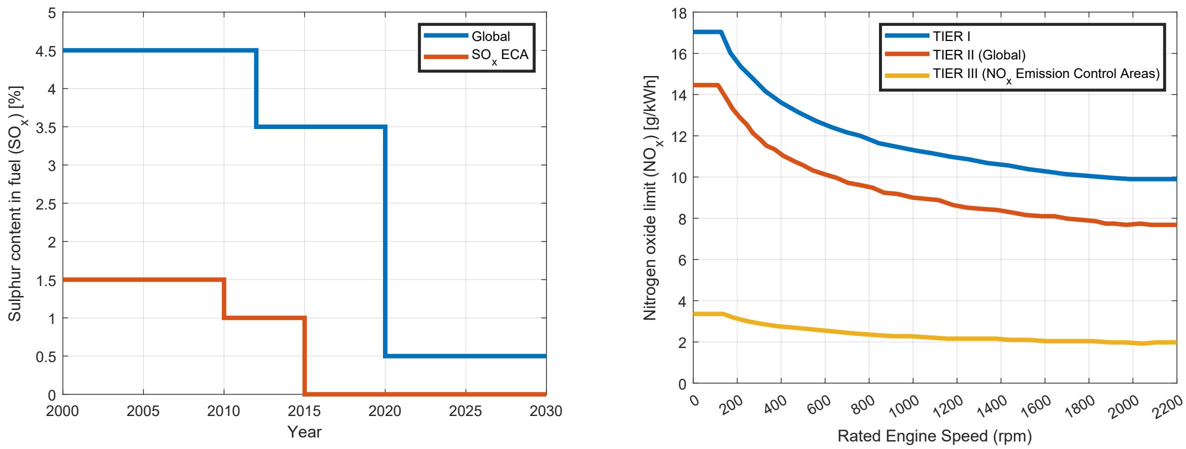


Figure 1.1: Global sources of greenhouse gas emissions and breakdown of CO_2 emissions from the energy sector and the transport subsector [1].

The International Maritime Organization (IMO) is implementing strategies to reduce the emissions from international shipping by establishing standards, imposing restrictions and provinding incentives. The initial strategy avoids a reduction in total GHG emissions from international shipping by at least 50% by 2030 compared to 2008, consistent with the Paris Agreement. IMO's current global standard for SO_x emissions is 0.5% mass by mass (m/m) by 2030 per the **International Convention for the Prevention of Pollution by Ships (MARPOL 73/78) Annex VI**, as shown in figure 1.2a. Similarly, the emission limit for NO_x from engines is dependent on the type of the engine and the nominal engine speed, as shown in figure 1.2b. The global limit has been set to 14.4 g/kWh for engines with speeds less than 130 rev/min and 7.7 g/kWh for engines with speed greater than 2000 rev/min.

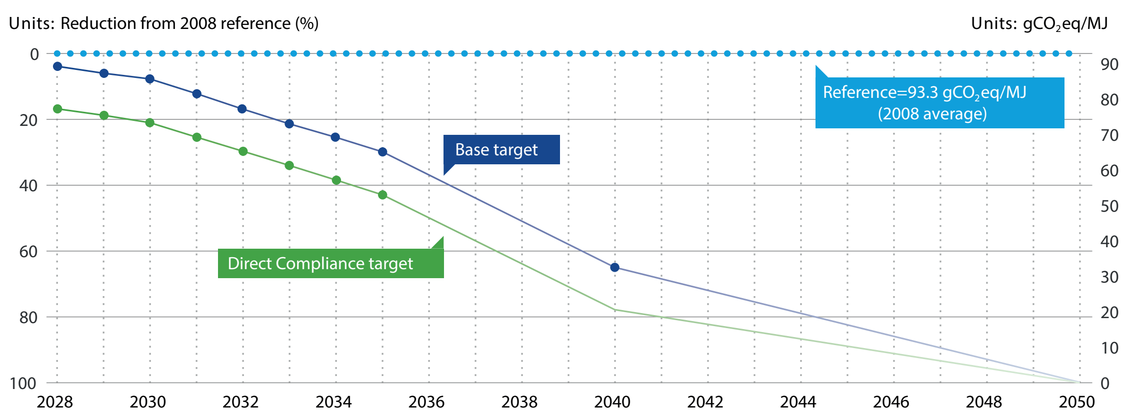


(a) SO_x limits.

(b) NO_x limits.

Figure 1.2: The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs) perspective.

With the **Net-Zero Famework (NZF)** approved bye the IMO in April 2025, the course is being set for the global maritime fuel transition [3]. The NZF aims to meet the emission reduction targets set out in the 2023 IMO GHG Stratey. Pending adoption in October 2025, a GHG fuel intensity (GFI) metric, in combination with a two-tier GHG pricing mechanism, will requiere operators to reduce their ship's GHG emissions intensity by 21% by 2030, with financial penalties for those who fail to do so. Beyond 2030, the requirements become rapidly stricter.



Year	2028	2029	2030	2031	2032	2033	2034	2035	…	2040
Base	4%	6%	8%	12.4%	16.8%	21.2%	25.6%	30%	…	65%
Direct	17%	19%	21%	25.4%	29.8%	34.2%	38.6%	43%	…	–

Figure 1.3: GFI reduction factors and reference value in NZF [3]. Two tiers of requirements are set on the annual attained GFI for a ship: a Base target and a Direct Compliance target. Each ship is required to meet the Direct Compliance target, either through the use of low-GHG fuels, or through one of the alternative compliance approaches. The Base target is used to separate between Tier 1 and Tier 2 compliance deficits.

Specificaly, The NZF aims to accelerate the adoption of low-GHG fuels and technologies, thereby supporting the achievement of the revised 2023 IMO GHG Strategy, namely a 20% reduction in emissions by 2030, a 70% reduction by 2040 (compared to 2008 levels), and net-zero emissions 'by or around' 2050, as is shown in fig 1.3. It is based on the GHG fuel intensity (GFI) metric expressed in gCO_2eq/MJ of all energy used on board in a calendar year on a well-to-wake (WtW) basis. Gradually stricter GFI targets are to be set every year from 2028. In effect, the NZF penalizes vessels with a GFI higher than the targets and incentivizes the use of low-GHG fuels and other technologies that can reduce the GFI. So, in response to the NZF, shipowners should carefully identify, evaluate and use technologies, fuels and solutions that help minimize energy consumption and lower GFI for ships.

Many ships contracted in the coming years may still be in operation in 2025, and need to consider the upcoming stringent requirements to retain their commercial attractiveness, asset value, and profitability in the following decades. In addition, beyond 2030, ships in operation may need to consider retrofit options to allow the use of low-GHG emission fuels.

The incentives to reduce emissions and improve the efficiency of equipment are motivating ship manufacturers to adopt zero-emission technologies. Fuel cells and energy storages (for example, batteries and supercapacitors) reduce GHG emissions and are the key enablers for the realization of low-emissions operation of marine vessels.

1.1.1 Propulsion Architectures

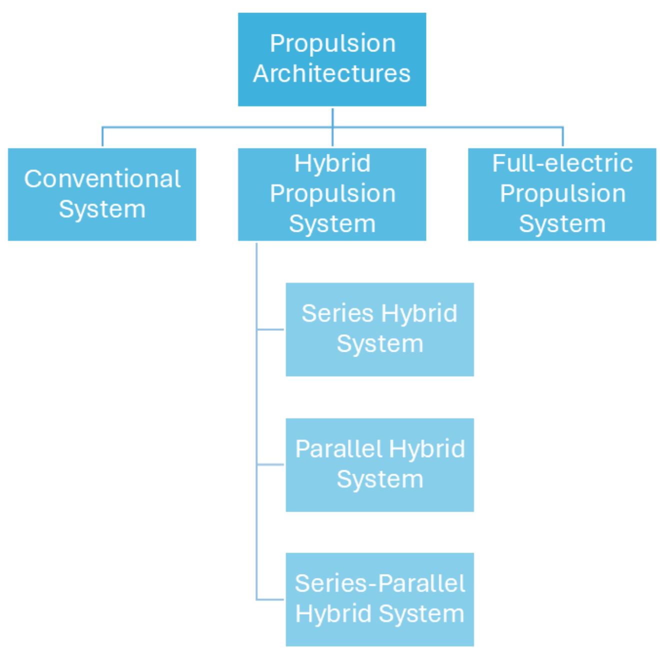


Figure 1.4: Types of propulsion systems−− > PDF

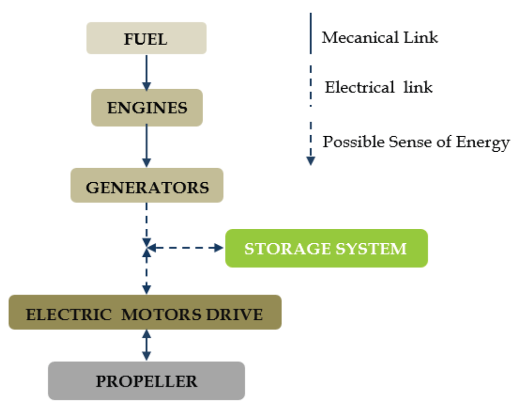
While conventional maritime transport is marked by the low technological development of the shipping and maritime industry, especially with regard to propulsion systems, which translates into a continuous increase in fossil fuel consumption, with the environmental impact that this represents, fully electric and hybrid propulsion architectures, made possible by innovative and high-performance energy storage systems (batteries, fuel cells, supercapacitors, etc.), are a promising way to significantly reduce fossil fuel consumption, pollutants, and greenhouse gas emissions [5].

New developments of electrically propelled ships are based on an integrated energy distribution system architecture that releases large propulsion machines [6].

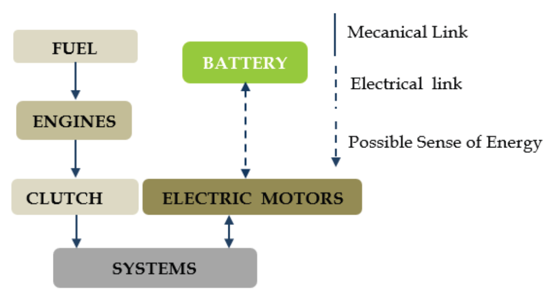
For full-electric propulsion, the main advantages include better dynamic performance (start-up, arrest, changes in speed, and regenerative braking), reduced fuel consumption due to the more efficient and flexible use of generators that work within their minimum specific fuel consumption region, and the possibility of having shorter shaft lines because only the electric motor must be installed close to the propeller.

The hybrid propulsion system uses a combination of diesel engines or gas turbines with electric motors to provide a wide range of propeller speeds [6]. One of the main advantages of this configuration concerns an increase in efficiency at low loads. Hence, less fuel consumption, fewer emissions, and a small footprint are achieved. Hybrid propulsion systems (HPS) are interesting when there is a large variation in power demand during operations. The design rules of this type of systems can differ significatly for different types of vessel, mainly due to the different types of mission profiles and operation requirements [5].
Hybridization of the propulsion chains are based on the separation or simultaneous use of several different energy sources. The choice of different power sources, and sources combinations is designed to adapt to the ship power and energy requirements which include the needs of low and high power, the needs of flexibility in changes of rhythm and the needs of redundancy and reliability.

As can be seen in the following subsections, there are three main architectures for hybrid propulsion chains [5]:



(a) Series Hybrid architecture [5].



(b) Parallel Hybrid architecture [5].

Figure 1.5: Semi-hybridization.

a) Series Hybrid Architectures

In this type of HPS architecture, two kind of energy system (combustion engines and electric systems) are used in series. The propulsion is done by a variable speed electrical drive (propulsion motor and power converter), as can be seen in the figure 1.5a. The electrical power of the propulsion motor is generally supplied by a set of generators driven by combustion engines.

Another option is to use fuel cells to provide electric power or a combination of fuel cells and generators driven by combustion engine. Energy storage systems (ESS) (battery, supercapacitors, etc.) can be connected to the electrical bus to provide propulsion power. This ESS can be used separately or with the generators. In this configuration, electric power is used as an energy vector between the combustion engine and the propulsion motor, so there will be no direct mechanical connection between the engine and the propeller. This configuration allows dissociating the operating points of the combustion engine and of the propeller. This is why it allows increasing the global efficiency of the system by optimizing the operating point of combustion engine and propeller in efficiency point of view. Of course, this advantage would be more significant if several generators and ESS are used, because it allows providing the required propulsion power by choosing an optimal combination of sources in order to keep the running combustion engines in their optimal operating area. However, for small ships, the possibility of multiplying the sources is very limited by volume and mass constraints.

b) Parallel Hybrid Architectures

Parallel hybrid architecture combines two kinds of propulsion motors: electric motors and combustion engines which are mechanically coupled by the same shafts through gearboxes and clutches, as can be seen in the figure 1.5b. The electrical motors and drive are supplied by electrical ESS and/or independent power sources generators driven by combustion engines and/or Fuel Cell for example. The two propulsion systems can be used together or separately. In most cases one of the propulsion systems is devoted to be used for low speed operations (classically the electric system) and the other one (classically the combustion engine) is used for high speeds. The first system can also be used as a boost system to provide a supplementary power during transient (starting, acceleration, etc.). This architecture reduces the number of elements compared to hybridization series and allows optimizing the sizing of each of the energy sources.

c) Series-Parallel Hybrid Architectures

The series-parallel architecture combines both architectures defined previosly. This association allows the passage of a parallel-type operation to a serial type operation.

Hybrid Propulsion systems are based on the combination of several power sources and energy storage systems.

1.1.2 Power Generation Systems

a) Steam Turbine

A steam turbine is a device that extracts thermal energy from pressurized steam and uses it to do mechanical work on a rotating output shaft [11] [12]. Steam turbines work by expanding steam through a series of blades mounted on a shaft, which turns a propeller to move the ship. They are particularly favoured for their smooth operation, high power output, and efficiency at high speeds, making them suitable for large vessels like liquefied natural gas (LNG) carriers and naval ships.

Steam turbines revolutionized ship propulsion with their introduction in the late 19th century, offering advantages such as small size, reduced maintenance, and low vibration. Despite the need for precise reduction gears or turbo-electric transmission to match the high RPM of turbines to the effective propeller speeds, the benefits included light weight and compact engine rooms.

b) Gas Turbine

Gas turbine engines are defined as aerospace engines that convert the chemical energy in fuel into thermal energy through combustion with air, resulting in the rapid expansion of gas that can perform work in the turbine and exhaust nozzle [13].

The gas turbine engine is recognized for its relative mechanical simplicity and its capacity to use a broad spectrum of fuels. The excellent power–weight ratio of gas turbine engines can be effectively exploited. The performance of gas turbine engines is affected by compressor efficiency, turbine efficiency, pressure ratio, peak cycle temperature, and combustion efficiency. Compressor and turbine efficiencies are continuously increasing through better design and manufacturing techniques, but little room is left for improvement. Increasing the pressure ratio can significantly improve the efficiency of an engine, but in order to retain sufficient power output, an increase in pressure ratio must be accompanied by an increase in maximum cycle temperature.

The first time a gas turbine was mounted on a watercraft occurred in 1947 in the United Kingdom, when a diesel free power turbine was mounted on a modified gunboat as a proof of concept [14]. This craft endured sea trials for 4 years, until the technology was proven to be a viable concept for the future of naval propulsion. HMS Grey Goose became the first fully gas turbine powered ship, being launched in the UK in 1953. It served in the British Navy for only four years; however, it has fulfilled its mission of proving the viability of gas turbine propulsion at sea. Since the inception of this technology, gas turbine engines or combinations of them with other engine technologies have been utilized in a wide range of watercraft, ranging from hovercraft all the way up to full-size aircraft carriers, with power outputs starting at single digits of MW all the way to hundreds of MW.

c) Diesel Engines

A diesel engine is a prime mover with the function to convert chemical energy stored in fuel to mechanical energy [5]. Diesel engine is a prime mover with the function to convert chemical energy stored in fuel to mechanical energy [7]. Compared to other prime movers such as a gas turbine or steam turbine, the diesel engine has a high efficiency. Its low fuel consumption is the main reason it is widely used in marine applications.

This power source is characterized by a high level of autonomy and by a relatively low cost compared to other sources. Diesel engines are characterized by a high level of greenhouse gases and pollutants emission particularly if they are used at a lower level of power than their rated power.

1.1.3 Energy Storage Systems

a) Supercapacitors

Supercapacitors can be interesting intermediate energy storage systems for small ships with short power transients [5]. They are characterized by a high value of charge and discharge peak power and can be used to rapid dynamic storage with very low specific energy. They are usually associated with slower systems, such as batteries and fuel cells, to manage high power short transients. They can operate at very low temperatures at which many types of electrochemical batteries stop working.

b) Flywheels

Flywheels can be used in some particular cases. Like supercapacitors, they are characterized by a high power density but a relatively low energy density. However using flywheels leads to implement safety systems to protect crew and passengers from possible flywheel failure. So their use in small ship context seems very limited.

c) Batteries

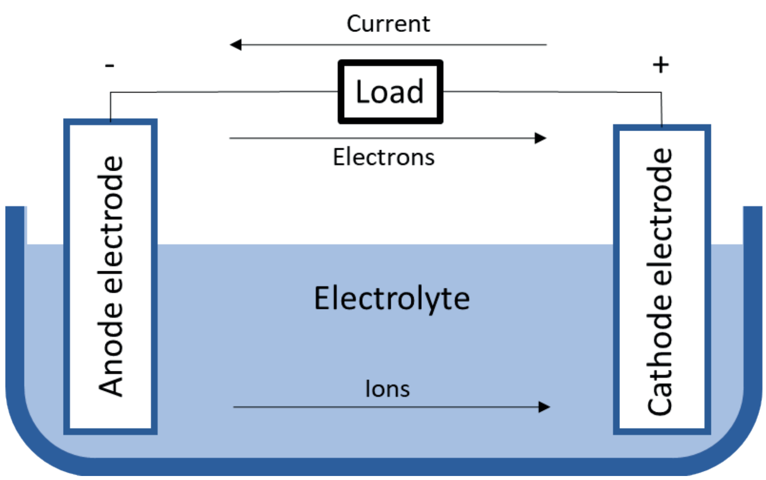


Figure 1.6: Components of a battery [19].

The oldest and most classical method for storing electrical energy is using batteries. In general, as can be seen in fig. 1.6, a battery is comprised of two different poles – a positive electrode called the cathode and a negative electrode called the anode. Then some material is used inside the battery, called electrolyte, that enables ions, or charge carriers, to be transferred back and forth between these poles by electrochemical reactions [19]. A separator can be placed between the cathode and anode, preventing them from touching each other. Hence, when the poles are connected by an electrical conducting material, electrons will flow through the external electrical circuit, and ions to flow through the electrolyte. This process allows energy to be stored or produced in the battery. The chosen energy carrier material, electrode and electrolyte composition, and the shape of the electrodes determine the properties of the batteries.

Battery technology has matured significantly during the past two decades as not only the world but also energy has become increasingly mobile [20].

Table 1.1: Energy Storage Systems Comparison [21].

Technology	Battery	Supercapacitor	Flywheel
Energy Density (Wh/kg)	50-250	0.05-5	5-100
Power Density (W/Kg)	50-2000	100k	1000
Lifecycle (cycle)	400-9000	50k	20k-100k
Lifetime (years)	5-10	5-8	15-20
Overall Efficiency (%)	85-99	60-65	15-20

As shown in the table 1.1, the battery provides the highest power density but lacks overall efficiency compared to energy storage systems. The flywheel, on the other hand, is the most durable system, offering a cycle life of between 20,000 and 100,000 cycles and a service life of between 15 and 20 years. However, due to its low power density, it may not provide enough power for a vessel.

The above comparison helps to understand why batteries are usually chosen over other types of energy storage systems: because of their reliability and high energy density.

1.1.4 Battery Systems in Maritime Applications

Battery application onboard ships can have multiple functional roles. While batteries can fully power a vessel for short distance or duration, improving performance and energy efficiency of the overall vessel is often the key purpose.

In 2018, the European Maritime Safety Agency (EMSA) commissioned DNV GL to perform a study on the use of electrical storage systems in shipping, with the objective of providing an overview of technology, research, feasibility, regulations and safety of battery systems in maritime applications [19]. Figure 1.7 shows the different uses of batteries recognized by EMSA:

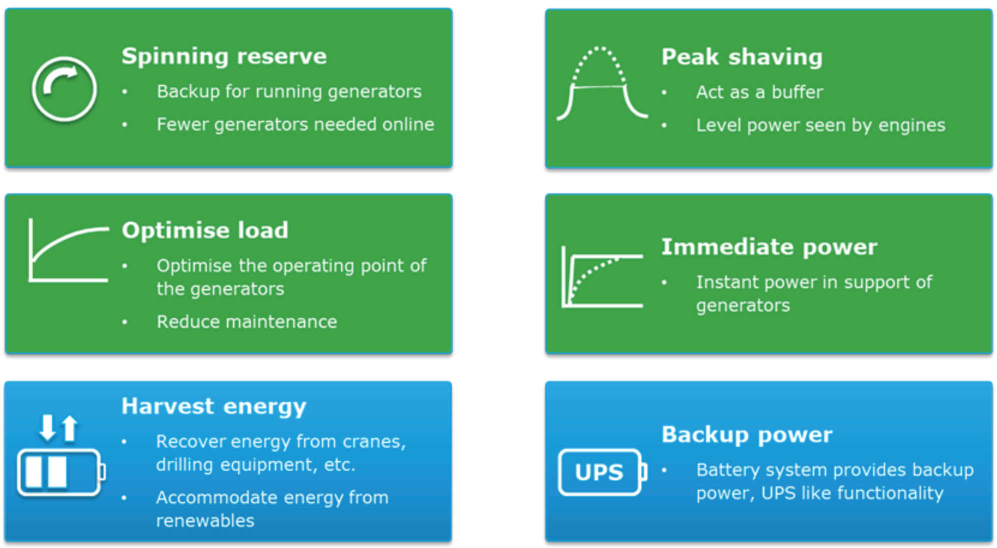


Figure 1.7: Funcitonal roles of battery systems onboard ships [19].

- **Spinning reserve:** This mainly applies to diesel electric dynamic positioning vessels. Using a battery-provided spinning reserve allows the vessel to reduce the number of generators running required to maintain the spinning reserve. It allows for lower fuel consumption, lower emissions, higher efficiency of running engines and lower maintenance costs due to reduced running hours [23].

- **Peak shaving:** Battery energy storage systems also provide additional electric power required for short-term additional power demands. In conventional propulsion systems, it is necessary to run additional diesel generator to meet those demands. Using a battery instead, the additional generator can be turned off, therefore saving on fuel and wear and tear of the engine. Vessels equipped with heavy electricity consumers such as cranes or electric-driven thrusters will benefit from this arrangement [23].

- **Optimise load:** BESSs are used to keep the load of the diesel engine/generator constant, allowing them to operate at maximum efficiency by charging when the load demand is low, then contributing to the fulfilment of the load request [25].

- **Immediate power:** This concept is recognized also like *Ramp Levelling* and is used to enhance dynamic engine performance [25]. Gas engines and other energy sources are not capable of handling fast load variations. In such cases, a BESS can be utilized to take care of the variations, while the main energy source produces slowly varying power [23].

- **Harvest energy:** This service consists of a time shift of energy generation by storing energy from photovoltaic panels, and/or other renewable sources such as wind, and then releasing it when they are absent. In addition, regenerative power improves ship energy management, using BESS to store energy obtained from the regenerative process of double speed engine braking (e.g., ships with cranes or large winches) [25].

- **Backup power:** BESS allows the vessel to stop diesel engines when onboard electricity consumption is low (e.g., in port). The generators are started only to charge the batteries. It is possible to reduce emissions because the diesel engines operate at optimal load (with maximum efficiency and minimum specific fuel consumption) during charging [23].

Additionally, other authors acknowledge the service of the **zero-emission mode** [25], that utilizes full electric propulsion for a limited time, reducing ship noise and vibration. Here, a BESS allows for switching-off all the onboard internal combustion engines. However, when fully powered by its own BESS, these ships are subject to specific technical limitations, such as max speed and maximum operation time, depending on battery power and energy content.

Among the commercially available battery technologies that can be used to provide services to boats, considering that is not feasible or possible to cover the entire spectrum of all technologies, because for any given chemistry there is most often a wide range of products representing different levels of quality and performance, information on some of them is provided [19]:

- **Lead-acid:** Lead-acid batteries are supplied worldwide by a large supplier base at a very low cost. Automotive batteries and industries where standby electrical power is critical are the biggest markets. It is considered to be very safe, since the electrolyte and active materials are not flammable; although the batteries are known to produce hydrogen under charging.The main drawbacks for such batteries are the specific energy and energy density. The power density is considered high for lead acid battery but is also here outperformed by Li-ion.

- **Nickel Cadmium (Ni-Cd):** Nickel cadmium batteries are mature technologies, that are widely used in UPS applications, for example. This technology is economically priced and presents the lowest per cycle cost. Good ventilation is important in the room to avoid explosive concentration of hydrogen. Ni-Cd have memory effect that causes a loss of capacity if not given a periodic full discharge cycle.

- **Nickel Metal Hybride (Ni-MH):** Nickel metal hybride has become one of the most readily available rechargeable batteries for consumer use and is used for the most at the same applications as Ni-Cd. It provides 40 percent higher specific energy than the standard Ni-Cd. Good ventilation is important in the room to avoid explosive concentration of hydrogen. The battery is more delicate and trickier to charge than Ni-Cd. High self-discharge is of ongoing concern, and devices with a Ni-MH battery gets “flat” when put away for only a few weeks.

- **Sodium Sulphur (Na-S):** The sodium sulphur batteries, are manufactured from cheap and plentiful raw materials and has higher specific energy compared to Li-ion batteries. However, the manufacturing processes and the need for insulation, heating and thermal management make these batteries quite expensive and counteracts the benefits, because these type of batteries need molten sodium-metal as a anode, making it necessary to operate at 300°C.

- **Lithium-ion (Li-ion):** Lithium-ion batteries can consist of different material and chemistries in the electrodes and the electrolyte, as well as manufacturing processes and related materials. Has the highest specific energy of commercially available batteries and highest energy density of commercially available batteries. Some disadvantages are flammable electrolyte and potentially limited availability of materials. Among the different **cathodes** and **anodes** of lithium-ion batteries are shown in the table 1.2:

Table 1.2: Lithuim-ion batteries types [19].

Lithium-ion Battery	
Cathode Chemistries	**Anode Chemistries**
Nickel Manganese Cobalt Oxide (NMC)	Graphene
Nickel Cobalt Aluminium (NCA)	Titanate
Lithium Cobalt Oxide (LCO)	Silicon
Lithium Manganese Oxide Spinel (LMO)	
Lithium Iron Phosphate (LFP)	

- **Li-polymer:** Also known like *solid-state battery*, this type of technology gives freedom in design of the battery geometry and improvement of the packing efficiency of the cells. It facilitates a long cycle life and offers the possibility of employing high-voltage cathodes. All these effects increase the practical battery energy density. The biggest challenge however is regarded as the volume changes of the electrodes during charging and discharging, that causes loss of contact between the electrode and the electrolyte. These effects all make ion conductivity low. An polymer interlayer between the electrodes and the solid electrolyte has been proposed to overcome these challenges. Some disadventagse of this technology are its high production cost and low lifetime.

- **Lithium-Sulphur (Li-S):** Like conventional lithium-ion batteries, these batteries use Li+ ions as energy carriers. They react with sulfur at the cathode giving some high theoretical values for specific energy and energy density. Hence these batteries will be very attractive for marine applications. Safety concerns related to these batteries, like dendrite formation, have been improved. However, this compromises the specific energy and energy density and increases the cost dramatically. Other obstacles to overcome are high electrical resistance, capacity fading, self-discharge, mainly due to the so-called shuttle effect.

About the different types of batteries described above, there is a technical comparation in the table 1.3.

Table 1.3: Energy Storage Systems Comparison [21]

Battery	Energy Density (Wh/kg)	Volumetric Energy Density (Wh/L)	Cost (USD/kWh)	Lifecycle	Efficiency (%)
Lead-Acid	20-40	60-110	165	500-1000	50-95
Ni-Cd	20-60	60-150	-	2000-2500	70-90
Ni-MH	50-70	140-300	146	500-2000	89-92
Na-S	~120	150-300	400	>1500	86-92
Li-ion	50-250	250-675	200-600	400-9000	85-99
Li-polymer	~150	250-730	-	-	-
Li-S	400	350	400-500	1500	-

Different ships have different operating profiles and batteries must respond to specific energy and power demand while also having in consideration the desired/expected life-cycle for the battery [19]. In this scenario, lithium batteries are superior to the other mentioned batteries in terms of energy density, lifecycle, and efficiency [21]. Lithium batteries have been gaining traction throughout the years, with many applications including portable electronics such as mobile phones, and automobile applications such as electric vehicles. Also are slowly being adopted for aerospace, military and marine applications. Lithium batteries are highly versatile energy storage devices that can be altered to meet specific demands, making them one of the best choices for marine application.

Considering the incorporation of batteries into the propulsion system of vessels, different propulsion architectures are recognized that provide a different use for batteries, even without considering them, as is the case with conventional propulsion architecture [19] [20]. This can be seen in the figures 1.8, 1.9 and 1.10.

a) Diesel-mechanic engine conventional system

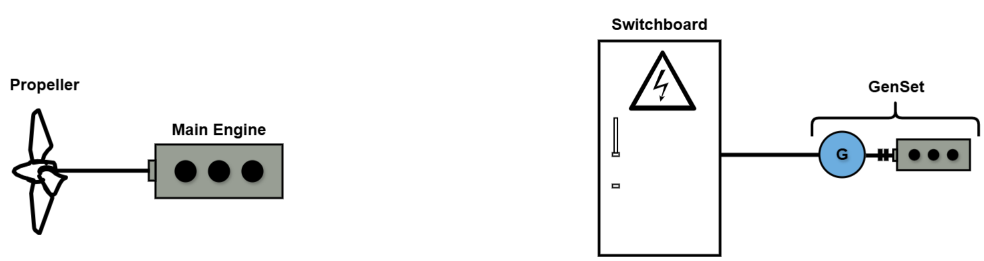
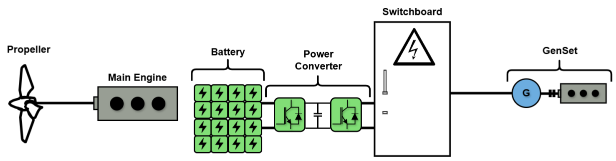


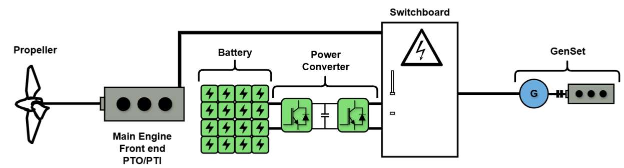
Figure 1.8: Traditional diesel-mechanic propulsion of a large merchant vessel −− > PDF.

In general, the configuration shown in Figure 1.8 is the most common installation on large ocean-going vessels [20]. It consists of a two-stroke main engine responsible for propulsion, while the ship's load is covered by auxiliary engines/sensets installed on board the vessel.

b) Diesel-mechanic engine hybrid system



(a) Semi-hybrid diesel-mechanic propulsion of a large merchant vessel with hybrid electric grid −− > PDF



(b) Full-hybrid diesel-mechanic propulsion of a large merchant vessel with front end PTO/PTI from RENK −− > PDF.

Figure 1.9: Semi-hybridization.

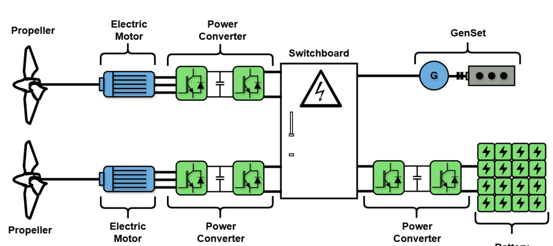
In Fig. 1.9a, a battery is included in a traditional system where the electric grid is separated from the propulsion of the vessel [20]. The battery will in such a solution be effective for smoothing the connected hotel electrical loads and contribute to handle large load steps or peaks [19]. When large load steps are increased, the number of auxiliary motors also increases. In some cases, the load can regenerate energy, for example, in crane operations. In such cases, the battery can be used to capture the energy.

In 1.9b, a battery is included in combination with a shaft generator power take-out (PTO) and power take-in (PTI) [20]:

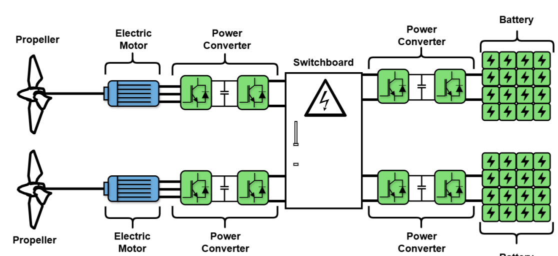
- **Power Take-Out (PTO):** The electrical/mechanical hybrid solution allows electricity to be generated by the main engine.
- **Power Take-In (PTI):** The electrical/mechanical hybrid solution allows propulsion power to be produced by generator sets and batteries. Additional power is possible (boost mode) when the main engine and PTI motor are running in parallel.

In addition, a ship with a hybrid electric/mechanical propulsion system may have a direct current distribution system, which can adjust the speed of the primary generator engines to the optimum fuel level depending on the load. This will reduce fuel consumption and minimize environmental impact [19].

c) Hybrid/electric propulsion system with electric motors



(a) Full-hybrid diesel-electric propulsion with batteries in a hybrid system −− > PDF



(b) Pure battery-electric propulsion −− > PDF.

Figure 1.10: Electric motor incorporation.

A full-hybrid diesel-electric solution, as is shown in fig. 1.10a, reduces the noise and vibration level on the ship [19]. Supplementary to be a source of energy for propulsion, the batteries will also smooth the load variations on the generator sets. This type of solution can for instance facilitate the use of zero emission operation when entering a harbour. Another version of this solution is with the batteries distributed into the drives directly [19]. In this way, each propulsion unit is independent of a common power source, which could be valuable for vessels that require highly reliable propulsion thrust.

In the fig. 1.10b, the propellers are connected to electric motors, which are driven by the energy stored in a battery system that is typically charged from shore [20]. In some battery-electric systems, a smaller diesel generator is sometimes included to ensure operation if the batteries fail to charge or to enable longer voyages, i.e. when the vessel has to dock [20]. For such a solution the batteries will be charged through an AC/DC converter. The converter can either be located on board the vessel or on shore. The fig. 1.10b shows two independent battery systems that deliver power to the thruster. This is according to class rules, which require that two independent battery systems are installed to provide propulsion power in case one of the systems fail [19].

Electrification of ship's propulsion is increasingly recognised as a core part of the maritime industry's future [23]. According to [26], the adoption of alternative fuels in the global fleet by number of ships is dominated by Liquefied Natural Gas (LNG) together with hybrid/battery ships:

- Of the total 1,349 alternative fuel ships in operation during 2022, there were:
  - 926 LNG ships.
  - 396 hybrid/battery-powered ships.
- Of the total 1,046 alternative fuel ships ordered, there were:
  - 534 LNG ships.
  - 417 hybrid/battery-powered ships.

It can be seen that the proportion of hybrid/battery propulsion systems has been increasing. However, it should be noted that the use of batteries as a source of propulsion for ships is limited to certain types of ships, the main ones being ferries, dynamic positioning ships and platforms, dredging, short-range ships, wind farm support vessels, and tugs. For ships that travel most of the time at constant speed, such as container ships, it make no sense to replace a diesel mechanical drive by hybrid concepts, because the diesel-mechanical drives are optimized for this operation and then getting a lower consumption than a diesel electric or hybrid system [4].

Harbour tugs are a sector of the maritime industry with great potential for the implementation of hybrid propulsion, as they experience significant variations in power demand during operations [5].

Ports are imposing more and more stringent regulations on exhaust and noise emissions. Those demands for cleaner operation can be easily met by the implementation of batery energy storage systems (BESS) (with hybrid-electric propulsion). Using a combination of electric power, BESS and combustion engines, a hybrid tug optimizes engine loading, resulting in lower specific fuel consumption, higher efficiency, lower emissions and lower fuel consumption.

1.1.5 Tugboats

Figure 1.11: Tugboat

Tugboats, as can be seen in fig. 2.2, are ideal candidates for hybridization due to their operational profiles, which involve prolonged idling, low-speed maneuvering, and frequent speed changes that lead to inefficient fuel consumption [21]. Among vessel types, tugboats—used for towing and maneuvering large ships—are among the highest emitters per unit of energy due to their highly variable load profiles [4].

A tug is a particular class of boat which helps mega-ships enter or leave a port. In addition to towing the vessel towards the harbour, tug boats can also be engaged to provide essentials, such as water, air, etc., to a vessel.

Low efficiency is further affected by the variability of the operating and load conditions to which engines are subjected. This variability leads to an increase in specific fuel consumption, which causes higher emissions and energy waste through exhaust gases and cooling systems [16]. However, this level of efficiency could be optimized by the hybridization.

1.2 Challenges and Research Opportunities

Several studies have examined hybrid propulsion systems [8], identifying the main topics of study related to hybridization, which are listed below [23]:

- Optimal sizing of the batteries.
- Optimal control strategy for the parallel operation of a battery and diesel generators.
- Control and safe operation of lithium battery packs.
- BESS control system architecture.

- Design of a battery management system (BMS).
- Safety assessment of the BESS, its elements and safeguarding measures required by the classification societies.
- Safety assessment of the BESS, its elements and safeguarding measures required by the classification societies.
- Degradation of Li-ion batteries.
- Battery application for large vessels.
- Reduction of fuel consumption and emissions.

Some studies related to hybridization are listed below, with emphasis on those related to tugboat hybridization:

- The feasibility of hybrid propulsion concepts is examined for different types of vessels, including tugboats [4], showing that there are definitely opportunities to save energy by using hybrid drive systems on ships, even return the much higher investment by savings in fuel.
- A genetic algorithm to obtain the optimal operating points regard with series hybrid configurations in power generation, especially for the case of heavy duty hybrid drive applications [18].
- A marine hybrid propulsion system, focusing on vector control of the electric motor during different modes and verifying the control feasibility [22].
- An analytical design methodology for BESS in hybrid marine vessels [24].
- The optimization of hybrid propulsion system design for a tugboat has been explored, presenting a methodology that streamlines powertrain component sizing and control, minimizing costs for a specific operating profile [27].
- A coordinated control strategy for a variable-speed hybrid tugboat have been presented to improve fuel economy. The proposed strategy, validated through simulations and a smallscale experimental testbed, showed reduced costs and lower CO2 emissions [28].

However, the challenge lies in finding a clear and detailed methodology for the hybridization of vessels such as tugboats that provides a path from the conceptual phase to the technical details and subsequent implementation, taking into account technical and economic aspects.

1.3 Thesis Objectives and Outline

The general objective of this work is to develop a methodology for the design and sizing of the battery bank for a hybrid tugboat, which will allow us to determine the benefits that would be obtained from hybridization.

Specific objectives include:

- Model the operation of a conventional tugboat
- Define hybrid architecture.

- Propose the service power of the electrical system.
- Preliminary design of the battery bank based on technical and economic criteria.
- Simulation of the proposed cell and battery bank.

1.4 Methodology

The methodology to follow for the development of this work is:

- **State of Art:** Review of existing hybrid propulsion system types and battery technologies.
- **Mathematical Analysis and Modeling:** Theoretical study of the tugboat's load profile and the implementation of batteries to support the vessel's operation.
- **Results:** Simulation of the models obtained to complete the conceptual analysis.

1.5 Scope and Limitations

The limitations of the work are:

- Architecture Layout.
- Bill of Components (BOM).
- Simulation considerations.

In conclusion, this work will contribute to the analysis of the different initial stages of hybridizing a tugboat, considering the essential technical aspects that allow us to understand the benefits of this conversion, with an emphasis on the electrical stage, specifically the batteries.

1.6 Chapter Summary

1.6.1 Chapter 2

1.6.2 Chapter 3

1.6.3 Chapter 4

1.6.4 Chapter 5

Chapter 2

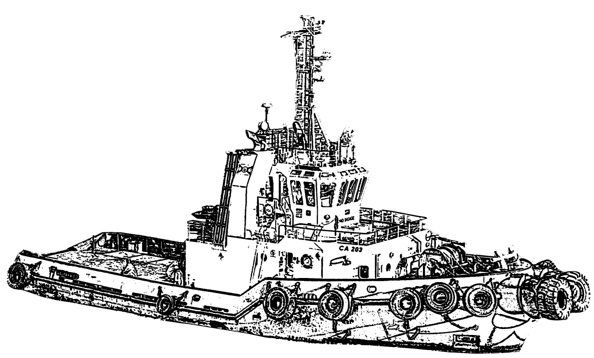
Tugboats: Description and Operation

A tug, or more commonly a tugboat, is an assistance boat which helps in the mooring or berthing operation of a ship by either towing or pushing a vessel towards the port [10] [15].

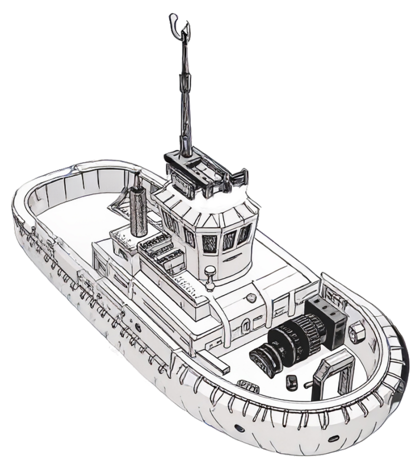
Tugboats are also essential for non-self-propelled barges, oil platforms, log rafts, etc. Due to their solid structural engineering, tugs are small but relatively powerful machines.

This chapter presents the technical and operational characteristics of tugboats, with the aim of identifying design aspects for the hybrid propulsion system and the sizing of the battery bank.

2.1 Overview of tugboats



(a) Harbour tug representation.



(b) Escort Tug representation.

Figure 2.1: Tugs representations.

A tugboat is technically defined as a type of vessel specifically designed to assist other ships in moving by providing a controlled amount of force. They are essential vessels in ports, as is shown in fig. 2.1, where they assist in the movement of floating objects and in emergency tasks [29].

In maritime law, maritime towing is defined as the operation whereby a vessel (the tugboat) transports another vessel or floating device that lacks self-propulsion or, despite having it, is unable to navigate under its own power, by pulling it through the sea. This functional duality—which encompasses both high-precision maneuvering assistance in congested environments (port towing) and long-distance transport and salvage (offshore towing)—requires design engineering and system redundancy adapted to very different operational risks [31].

The importance of tugboats has grown exponentially with the phenomenon of naval gigantism [32]. Large modern ships, such as cruise ships and oil tankers, have tremendous momentum (inertia) that prevents them from stopping quickly or maneuvering effectively on their own in narrow channels or docks [30].

Therefore, tugboats are essential to counteract these inertial forces. Support is necessary for entry, exit, and general maneuvering, including assisting ships in emergency situations [31].

Technological innovation plays a key role in tugboat operations. Environmental challenges and high efficiency requirements have driven the digitization and development of more sustainable and less polluting tugboats [32]. This trend includes the integration of automatic control technologies to optimize maneuver management, such as investment in hybrid and electric propulsion systems.

The efficiency of these types of vessels is based on fuel consumption in relation to load capacities, always taking into account both the propulsion system and the energy conversion system that determines it. The efficiency of conventional propulsion systems varies between 38% and 41% for a power range between 100 kW and 2,500 kW. For vessels with higher power ratings, efficiency increases to 48% [16] [17].

2.1.1 Types of tugboats

The classification of tugboats is primarily based on their operational domain and assigned primary mission. The usage and functions of tugs vary from port to port, as different ports have different requirements and intakes. The common thing is pushing or towing mega boats or barges. Their usage depends on the following factors:

- Port traffic Volume,
- Types of ships to be served by that tug,
- Navigational obstacles to be catered to,
- Conditions of environmental protection,
- Local laws and
- Domains to be carried by a tug.

a) Harbour Tugs

Figure 2.2: A harbour tug in the port of Algeciras [34].

These vessels are specialized in assisting large ships entering and leaving port limits, as can be seen in fig. 2.2. Their design prioritizes agility and the ability to apply force in any direction with maximum efficiency [29].

- **Design:** Typical dimensions vary in length between 20 and 30 meters, with drafts of 3.0 to 4.5 meters. Their empty displacement speed ranges from 5 to 13 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 400 - 3,000 hp (300 and 2200 kW) [33].
- **Propulsion:** Their designs have reduced length and increased beam and depth in order to install greater propulsion power, variable pitch propellers, Kort-type rotating nozzles with fixed rudders or rudder flaps, or Tow-Master systems (fixed nozzles with five rudders, two at the bow and three at the stern) to achieve greater traction and maneuverability [33] [37].

b) Offshore/SAR Tugs

Figure 2.3: SAR BALDER, an offshore Tug built in 2006 [36].

An offshore or *search and rescue* (SAR) Tug, is designed to operate in the open sea, these tugboats are characterized by their robustness, greater autonomy, and ability to withstand severe ocean conditions. An example of this vessel is shown in fig. 2.3.

- **Design:** Typical dimensions vary in length between 25 and 70 meters, with drafts of 4.5 to 6 meters. Their empty displacement speed ranges from 5 to 13 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 1500 - 6,000 hp (1120 - 4470 kW) [33].

- **Multifunctionality:** In addition to long-distance towing tasks, they are equipped for rescues and complementary firefighting and pollution control tasks [38]. An example of a modern rescue vessel is 40 meters long, has 5,300 hp and a Bollard Pull of 60 tons [39]. These vessels are equipped to provide any type of assistance to deep-draft vessels [38].

c) Escort Tugs

Figure 2.4: Robert Allan Rastar 3200 Escort Tug [35].

These tugboats perform a critical safety mission. Their main objective is to provide preventive escort to ships carrying hazardous materials (such as LNG or crude oil) in high-risk areas, standing ready to counteract any possible loss of control of the assisted vessel.

- **Design:** Typical dimensions vary in length between 40 and 80 meters, with drafts of 5 to 6 meters. Their empty displacement speed ranges from 15 to 16 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 6,000 -20,000 hp (4470 - 14,900 kW) [33].
- **Technology and Bollard Pull:** They require extremely robust and powerful designs to maintain control of a vessel moving at cruising speeds. An example of a modern escort tugboat is shown in figure 2.4, features Rolls Royce azimuth thrusters and is designed to generate fixed-point traction exceeding 85 tons [40]. They are equipped with constant tension maneuvering winches to maximize safety during operations.

d) Tugs Comparation

Below, in the table 2.1, there is a summary of key operational and dimensional characteristics:

Table 2.1: Operational Classification and Dimensional Parameters of the Tugboat

Type of Tugboat	Operation	Design objective	Typical BP
Harbour Tug	Protected maritime and port areas such as bays or estuaries.	Maneuverability, Reduced Draft	40-80
Offshore/SAR Tugs	Open Ocean	Autonomy, Stability, SAR Power	60-200+
Escort Tugs	High-Risk Areas (LNG, Crude Oil)	Rollover Control, Side Pull Capacity	60-120

2.2 Tugboat Sizing Characteristics

2.2.1 Dimensions

a) Length Overall

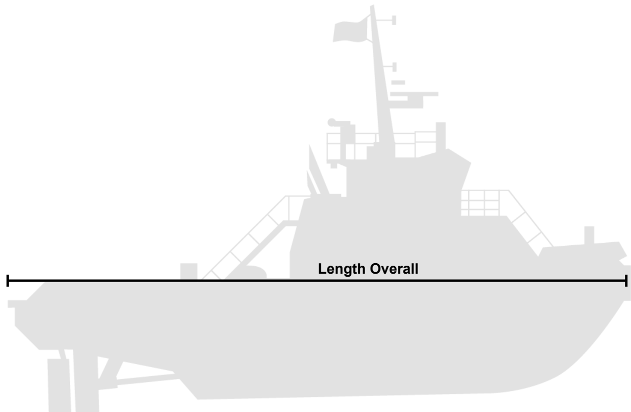


Figure 2.5: Length Overall

The overall length is measured from the bow to the stern of the vessel in a straight line, excluding bowsprits, rudders, outboard motors and their supports, cranks and other accessories, fixed elements, and extensions [43].

b) Beam

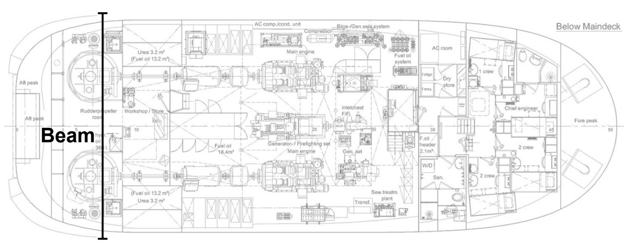


Figure 2.6: Beam Tug

A tugboat's beam is the distance between the two widest points of the hull [44]. It is usually measured at the waterline, although other specifications such as maximum beam or deck beam may be used. Beam is a critical dimension in naval architecture, affecting the stability, maneuverability, and overall performance of the vessel [45].

c) Draught

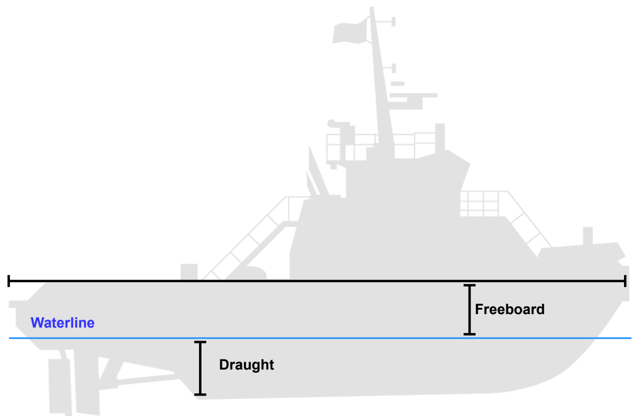


Figure 2.7: Draught Tug

Draught or Draft is the vertical distance from the moulded base line amidships to the actual waterline [11].

d) Depth

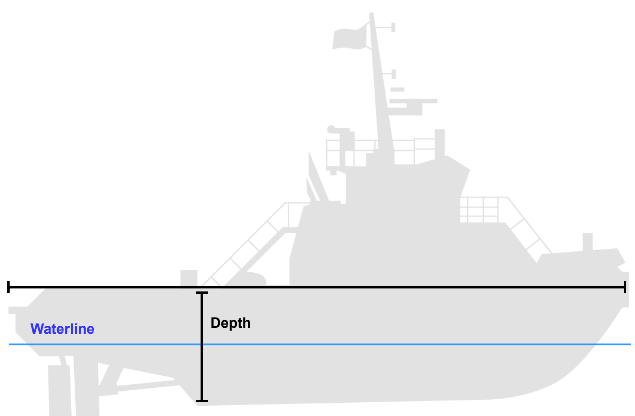


Figure 2.8: Depth Tug

Depth or molded depth of a vessel is measured at the center of the length, from the top of the keel to the highest continuous deck on the side [11].

2.2.2 Volumen and Weight

a) Gross Tonnage (GT)

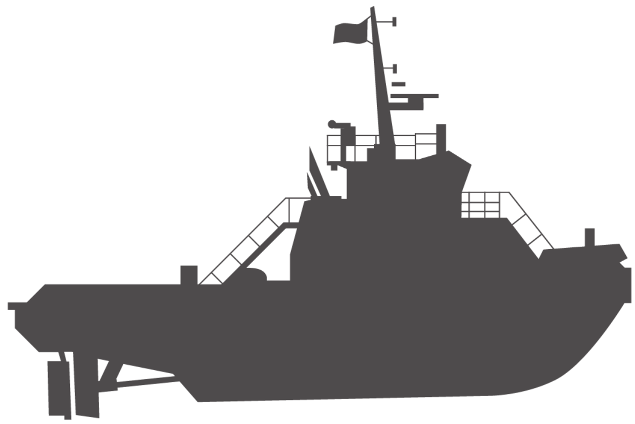


Figure 2.9: Gross Tonnage Tug

Gross Tonnage (GT) measures the “total inside space” of the ship, including all areas like cargo holds, engine rooms, and crew spaces. It’s about “volume”, not weight [11] [46].

- Under vessel measurement rules of various nations, the Panama Canal and the Suez Canal gross tonnage is a measure of the internal volume of space within a vessel in which 1 ton is equivalent to 2.83 cub m or 100 cub ft.
- Under the International Convention on Tonnage Measurement of Ships, 1969, (ICTM), a standardized numerical value is a logarithmic function of spaces within a vessel. There is no definition of a ton under ICTM because the value per unit of volume is greater on a vessel of large volume than on a vessel of small volume. Gross tonnage according to the national and canal rules generally includes spaces bounded by the under surface of the uppermost complete deck, the side frames, and the floor frames or the inner bottom if it rests on the floors or if the double water is for water ballast, plus closed-in space in deck structures are available for cargo or stores or for the berthing or accommodation of passengers or crew. Rules vary greatly as to exclusion or inclusion of various spaces. Gross tonnage according to ICTM is GT = K1 V in which V is the total moulded volume of all enclosed spaces of the ship in cub m and K1 is 0.2 + 0.02 log10 V.

b) Net Tonnage (NT)

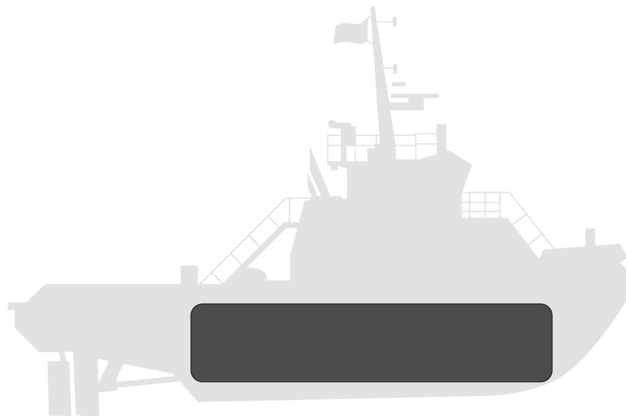


Figure 2.10: Net Tonnage Tug

Net Tonnage (NT) is focuses on the space available for cargo [11] [46].

- Net tonnage according to the national and canal rules is derived from gross tonnage by deducting an allowance for the propelling machinery space and certain other spaces.
- Net tonnage according to ICTM is a logarithmic function of the volume of cargo space, the draft-to-depth ratio, the number of passengers to be accommodated, and the gross tonnage.

c) Deadweight (DWT)

DWT refers to weight and indicates how much the ship can safely carry. It is expressed in long tons of 2,240 pounds (1,016.0469088 kilograms) [47].

d) Displacement Tonnage

Displacement tonnage is used to define the size of naval ships. It refers to the weight of the volume of water displaced by a vessel in normal seagoing condition [47].

2.3 Technical Characteristics of the Propulsion System

2.3.1 Bollard Pull - BP

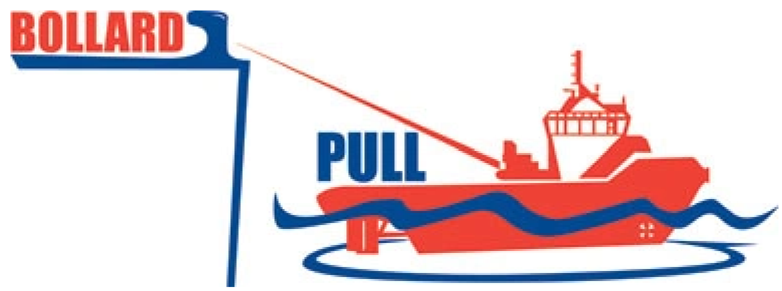


Figure 2.11: The bollard pull test measures a boat's maximum towing capacity by securing it to a fixed point and measuring the force generated at full thrust while maintaining zero forward speed [30].

Bollard pull (BP) is the fundamental metric for quantifying a tugboat's pulling power.

BP measures the maximum force a vessel can exert while stationary in the water. It is the maritime equivalent of horsepower or torque in land vehicles. The measurement requires securing the vessel to a fixed bollard at the dock and using specialized load cells to calculate the tension generated on the line with the engines running at full power, as is shown in the figure 2.11.

BP specifications are expressed in units such as kilonewtons (kN), short tons (stf), or ton-force (tf) and are critical for demonstrating the vessel's pulling capacity under load. All commercial vessels intended for towing operations must obtain fixed point pulling certification from international classification societies such as the American Bureau of Shipping (ABS) or Bureau Veritas (BV) [32].

The tugboat's design is specifically geared toward maximizing pulling force in relation to its size. A common and effective way to increase pulling force is by adding a nozzle (duct) that surrounds the propeller, which improves propulsive efficiency [32].

2.3.2 Propulsion Engineering for Maneuvering

The maneuverability of a tugboat, which directly influences its operational efficiency and safety, is intrinsically linked to the design of its propulsion system.

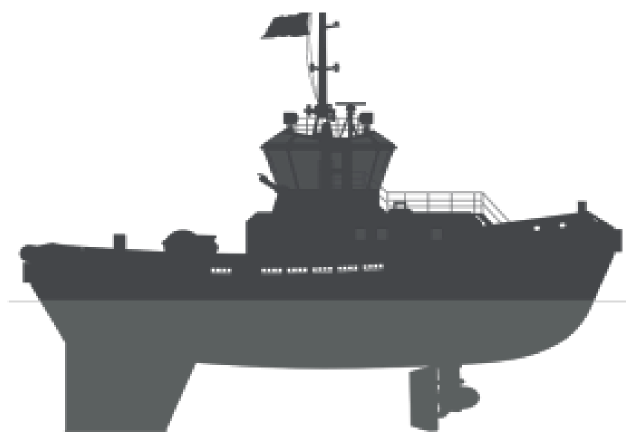
a) Conventional Propulsion

Figure 2.12: Conventional propulsion: A tug fitted with fixed propellers, single or twin screw (left or right handed) and single rudders with fixed nozzles [48].

It uses a standard diesel engine connected to a propeller, as is shown in the figure 2.12. It is a proven and simple system, with good efficiency in linear towing tasks within ports and coastal areas. However, it offers limited lateral maneuverability compared to more advanced systems [41].

b) Azimuthal Stern Drive (ASD)

(a) Azimuth Stern Drive (ASD): A tug fitted with two azimuth thrusters in nozzles at the stern, capable to direct all its thrust in any direction [48].



(b) ASD Tractor: A tug with azimuth thrusters located generally forward off the midship and a rudder-shaped fin at aft side. [48].

Figure 2.13: ASD Tugs representations.

These are rotary thrusters, where a motor is capable of directing all its thrust in any direction [41]. They all have two propellers, and since each one can be oriented independently, maneuverability is greatly increased, allowing them to turn on themselves in a single length [42], as is shown in the figure 2.13a. The only difference with conventional tugboats are the modifications to the stern and structure to house the propellers and withstand the forces exerted by the nozzle rudder, which are different from those exerted by a conventional rudder and propeller [42].

The in-line propulsion propeller (Z-Drive) is a system similar to the azimuth propulsion system, but with the difference that the propeller is oriented in only one direction (usually straight ahead), although it still provides great maneuverability in towing operations. In terms of power transmission, these tugs are more efficient [41].

An ASD Tractor Tug generally has two fixed pitch azimuth propellers in the center of the boat or slightly forward of the center of the boat, as is shown in the figure 2.13b.

The propulsion units are located well forward of the tow point, close to the pivot point, which generates a large turning thrust. This can result in poor steering performance, which is remedied by installing a large center rudder-shaped fin at aft side [49].

c) Voith-Schneider Propulsion (VSP)

Figure 2.14: Voith-Schneider Propeller (VSP) propulsion: A tug with VSP generally located forward off the midship and a rudder-shaped fin at aft side [48].

This system, as is shown in the figure 2.14, offers omnidirectional thrust and total steering control, resulting in formidable positioning accuracy. These harbor tugs have been equipped with *VoithSchneider* propellers, a unique type of propulsion system consisting of oscillating blades that allow them to have total, targeted control over the vessel without having to adjust the orientation of the main engine or propellers [41].

This propulsion system allows for highly precise operation in any direction due to its agility, making it ideal for rescue operations as well as in ports crowded with high-density maritime traffic [41].

d) Propulsion Comparation

The transition from a conventional system to an azimuth system, and even more so to a Voith-Schneider system, involves an increase in engineering complexity and acquisition cost, but this is fully justified by the need to operate with high precision and safety in confined spaces or under adverse weather conditions. The design of these advanced systems is crucial for safety, as they allow the tugboat to apply the required force (BP) without compromising lateral stability and minimizing the risk of capsizing.

Table 2.2 compares the different types of propulsion discussed in this document.

Table 2.2: Comparison of Propulsion Systems in Tugboats and Performance.

Propulsion System	Advantages	Maneuverability	Typical Application
Conventional	Low Cost, Proven Design	Limited (Good in a straight line)	Single Coastal/Offshore Trailer
ASD	High BP, Axis Rotation, Port Agility	Good	Port Towing and Escort
VSP	Omnidirectional Thrust, Extreme Stability	Very Good	Congested Ports

2.3.3 Main Diesel Engine

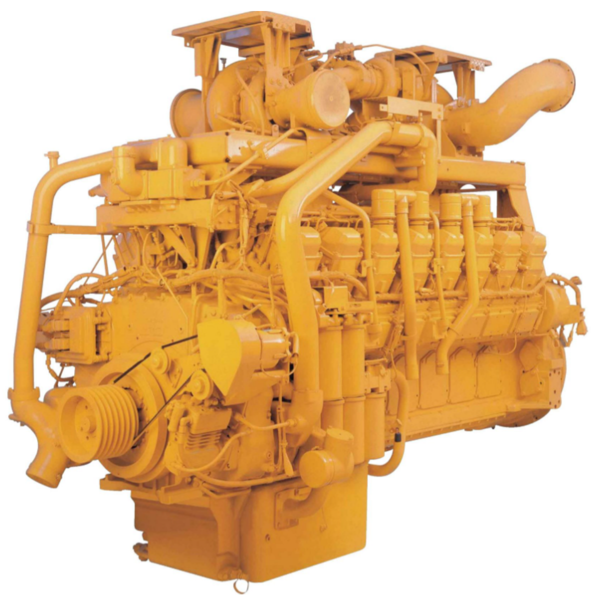


Figure 2.15: Main diesel engine for a tugboat propulsion system.

Diesel engines for propulsion, as is shown in figure 2.15, are designed to increase efficiency and keep costs to a minimum, whether in port or at a terminal. They are characterized by high power density, acceleration, and static traction.

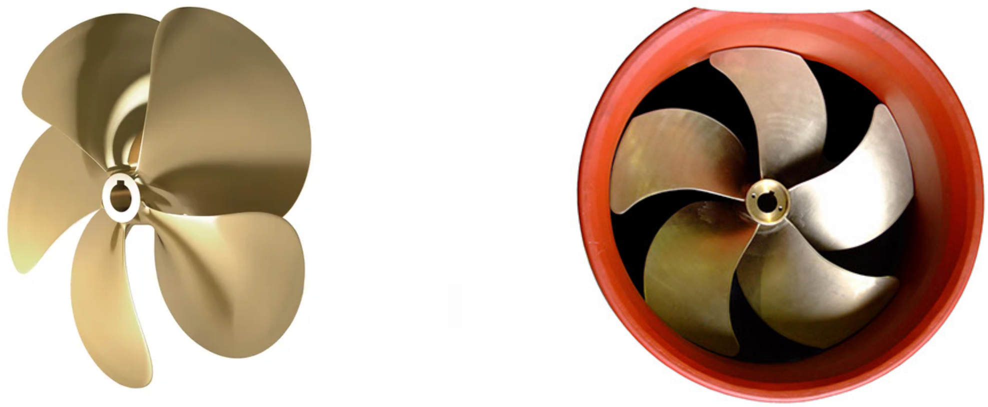
In general, the output power is available with variable load for an unlimited time, with an overload capacity of 10% of the rated power, for use in emergencies for a maximum period of time (1 out of every 12 hours, for example), considering that overload operation cannot exceed a certain number of hours per year.

Regard with the main diesel engines for propulsion system, the brake horsepower (BHP) is the power measured directly at the engine of a boat before any power loss through the transmission system or propeller shaft [50]. It influences maneuverability when docking and handling in adverse conditions.

BHP provides a realistic figure for useful power, as any kinetic application, such as a boat, will suffer a certain degree of resistance from the sea, which will reduce the actual power. Therefore, BHP can be thought of as the gross power of the engine, while the actual useful power will be somewhat lower due to mechanical losses in the transmission.

2.3.4 Conventional Propellers

a) Fixed Pitch Propellers



(a) Fixed pitch propeller.

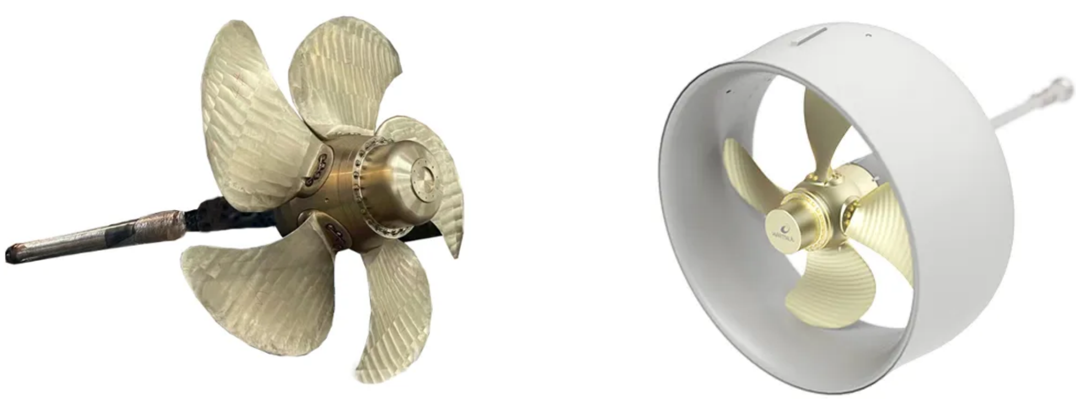
(b) Fixed pitch propeller with nozzle.

Figure 2.16: Fixed pitch propellers configuration.

These types of propellers, as is shown in the figure 2.16a, maintain their configuration unchanged, meaning that their blades do not move on their axis [33].

Also, there are fixed pitch propellers with improved propulsion efficiency thanks to the nozzle that aids vector thrust [33], as is shown in the figure 2.16b.

b) Variable Pitch Propellers



(a) Variable pitch propeller.

(b) Variable pitch propeller with nozzle.

Figure 2.17: Variable (controllable) pitch propellers configuration.

Unlike fixed-pitch propellers, in a variable pitch propeller (also known like controllable pitch proppeller) each of the blades can rotate on its own axis [33]. An example can be seen in the figure 2.17a, where each blade is attached to the central shaft by means of an anchor that allows it to rotate.

In some cases, the propulsion efficiency can be improved thanks to the nozzle that aids vector thrust [33], as is shown in the figure 2.17b.

e) Propellers Comparation

Variable pitch propellers are more efficient than fixed pitch propellers because the adjustment of the blades allows for maximum power or variable speed, which is not the case with fixed pitch propellers that are designed for specific ordinary operating conditions.

It should be noted that variable pitch propellers do not provide as much thrust for reverse navigation as fixed pitch propellers do, which limits the tasks that tugboats can perform.

2.3.5 Special Propellers

a) Schottel system

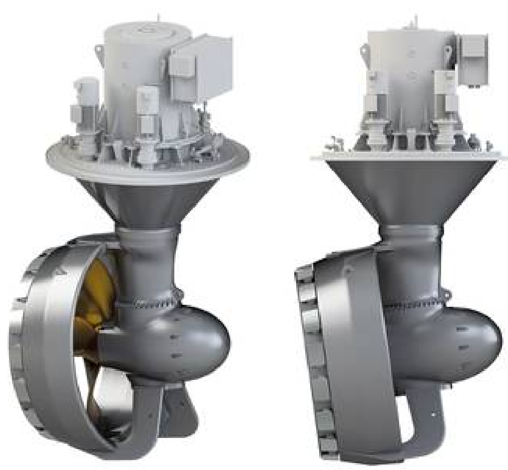


Figure 2.18: Schottel system.

In this system, the propeller positioned at a 90º angle to the vertical, fixed to the shaft is a nozzle inside which the propeller rotates, as is shown in the figure 2.18. The entire assembly can rotate 360º on the vertical axis, providing excellent maneuverability and allowing movement in any direction.

b) Voith-Schneider System (Cycloidal Propeller)

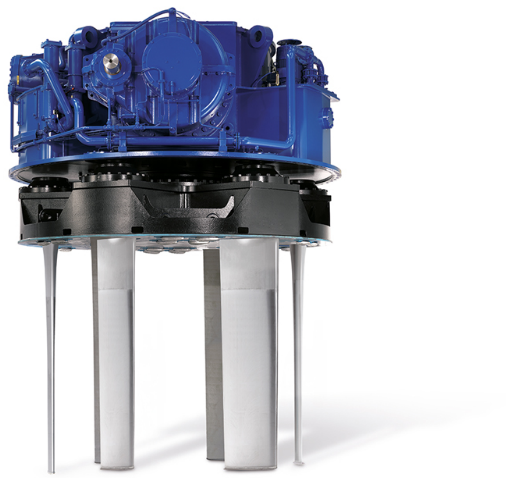


Figure 2.19: VSP propeller.

This propeller system consists of a rotor that rotates on a vertical shaft fixed to the hull approximately at its pivot point, as is shown in the figure 2.19. It is equipped with several blades that pivot on vertical axes controlled from the bridge by a device called a steering control. This device is responsible for directing the thrust in the opposite direction to the desired thrust. The mechanism is synchronized to center the thrust at a single point.

2.4 Naval Maneuvers and Assistance Techniques

Towing operations are governed by strict procedures that seek to maximize safety in dynamic and often restricted environments.

2.4.1 Docking and Undocking Procedures

The assistance of tugboats is crucial during docking and undocking maneuvers. The undocking maneuver can be as laborious as the docking maneuver and must be planned in advance. The objective is to leave the assisted vessel clear of the dock, in a safe position that allows the rudder and propeller to be effective so that the vessel can gain control and momentum.

a) Operational Coordination

The maneuver is initiated by request via VHF radio. The bow and stern operating personnel receive instructions to unmoor the lines.

b) External Agent Management

It is essential to consider the effects of winds and currents. Strong winds from outside can add to the effects of the engine in reverse during undocking, creating a dangerous situation that can violently push the stern or bow toward the dock. The ability of the highly maneuverable tugboat to apply instantaneous lateral force reduces the time the assisted vessel is exposed to the combined and unfavorable action of wind and propulsion, mitigating the risk of collision. If the wind action is severe, tugboat assistance is mandatory, especially if the actions of the assisted vessel's capstan are insufficient.

2.4.2 Tugboat Performance

The assistance of tugboats is crucial during docking and undocking maneuvers. The undocking maneuver can be as laborious as the docking maneuver and must be planned in advance. The objective is to leave the assisted vessel clear of the dock, in a safe position that allows the rudder and propeller to be effective so that the vessel can gain control and momentum.

Tugboats use various configurations to apply controlled force:

a) Tugboat Working on a Beam or Over a Cable

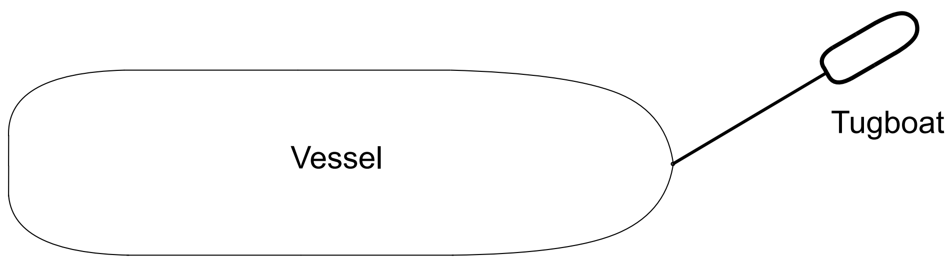


Figure 2.20: Arrow maneuver or rope maneuver [33].

In this procedure, the tugboat works separately from the vessel it is assisting, pulling it from the end of a rope, which can be attached at different points on the vessel, thus performing various functions (towing, holding, etc.), as is shwon in the figure 2.20. This procedure prevents direct contact between the two vessels and also ensures that all the power of the tugboat is exerted in the direction of the rope. The disadvantage of this procedure is that it requires more maneuvering space due to the length of the mooring line, so the system cannot be used where space is limited [33].

b) Bow-Supported Tugboat (Ram Locking)

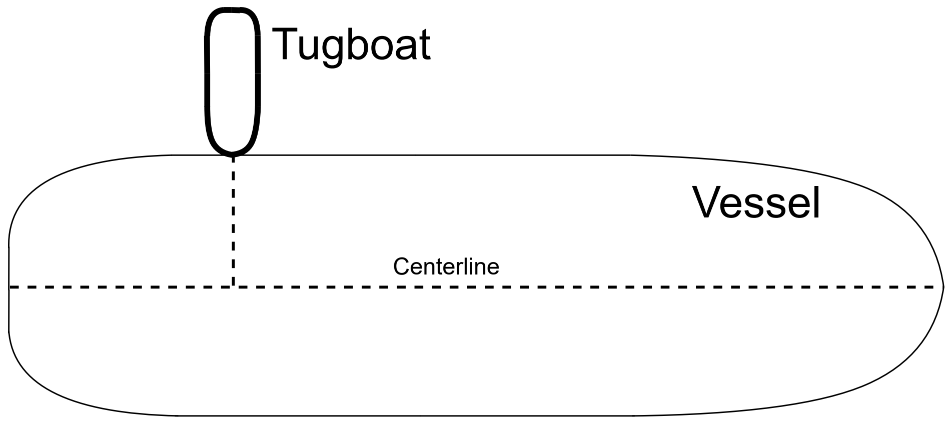


Figure 2.21: Maneuver supported by the bow [33].

In this procedure, the tugboat rests its bow on the side of the vessel it is assisting and pushes it in a direction that is essentially perpendicular to the centerline (the straight line running from stern to bow), as is shwon in the figure 2.21. It is common in this procedure for the tugboat to be secured to the ship with 1, 2, and 3 mooring lines, which prevents relative slippage between the two vessels during the maneuver [33].

c) Tugboat alongside

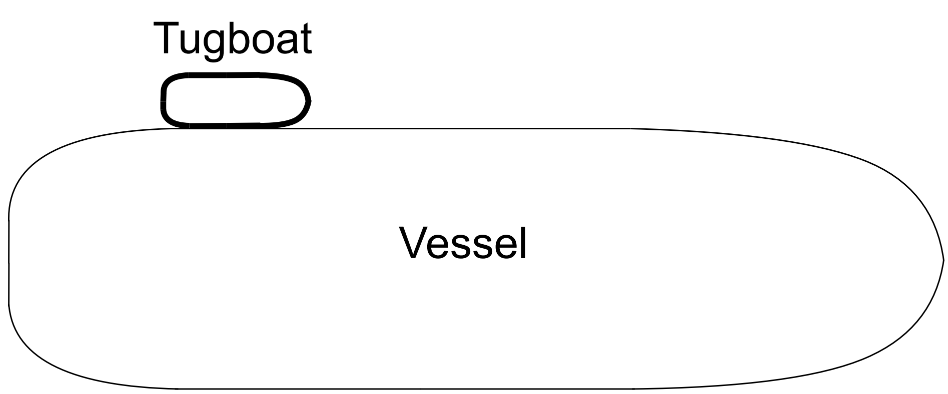


Figure 2.22: Alongside Maneuver [33].

In this procedure, the tugboat is positioned alongside the ship and roughly parallel to it, moored to the ship by several ropes that ensure the transmission of force, as is shown in the figura 2.22. This procedure is generally used to maneuver ships that do not have sufficient propulsion, in confined spaces and in very calm waters [33].

2.5 Operational Profile

Figure 2.23: TUG route to the Puerto Montt's pier.

Next, we will analyze the operational profile of a tugboat in Chile, considering that it has an operating schedule of 10 hours of work, performing this task in Puerto Montt. Figure 2.23 shows the route taken by a tugboat from the shipyard to the dock in Puerto Montt, working in Flecha or Sobre Cabo mode or performing a alongside maneuver, assisting ships that need to enter the port or leave the dock to go out to sea. The route taken by the tugboat is divided into three different sections.

Table 2.3: Dimensions

Dimensions Tugboat	
Length overall	25 m
Beam	10.5 m
Depth	4.6 m
Draught	3.9 m
Gross Tonnage	269 tons
Deadweight	220 tons

Table 2.4: Machinery

Machinery Tugboat	
Main Engines	2 x Cat 3516 B
Total BHP	4894 HP (3650 kW)
Bollard Pull	65.4 tons
Propulsion	2 x Schottel SRP 1215

- **Blue:** Low-speed route, tugboat departure from shipyard.
- **Yellow:** Nominal propulsion speed, tugboat moves toward the dock to perform operational maneuvers.
- **Red:** High-power section, operational maneuver performed by a tugboat assisting a ship.

Based on the route to be analyzed and the maneuvers that the tugboat must perform at the dock, there are two operating profiles for each maneuver, as it can see in the figure 2.24 and figure 2.25.

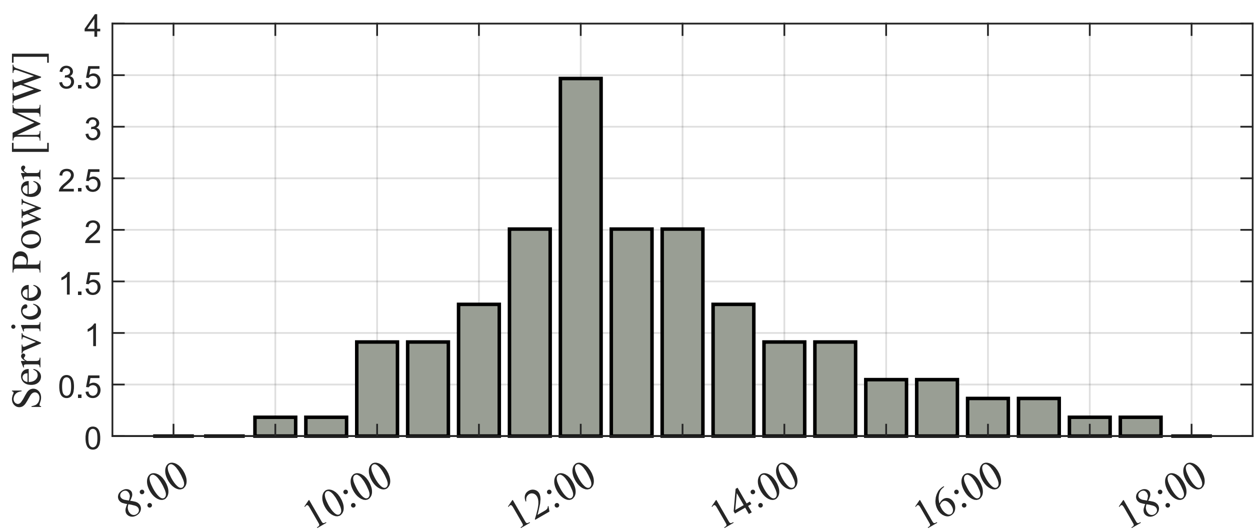


Figure 2.24: Operational profile: one maneuver.

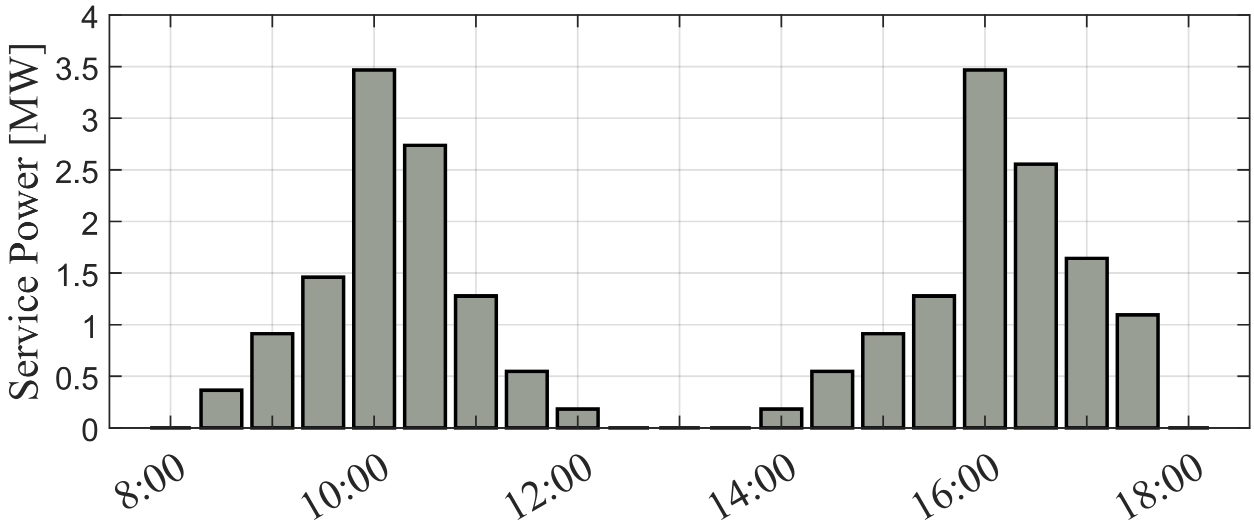


Figure 2.25: Operational profile: Two maneuvers.

2.5.1 Brake-Specific Fuel Consumption (BSFC)

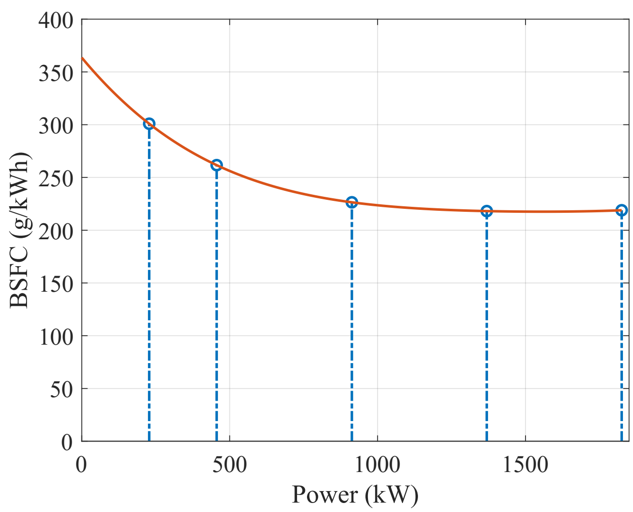


Figure 2.26: Brake-specific fuel consumption of a diesel engine model Cat 3516 B.

To analyze the operating profile, it is necessary to introduce the concept of **brake-specific fuel consumption (BSFC)**. BSFC is a measure of an engine's fuel efficiency, calculated as the amount of fuel consumed per unit of power produced over a specific time [8].

The BSFC as a function of output power is illustrated in fig. 2.26 for a typical engine used in tugboats. At higher power levels, the BSFC is lower, indicating that more power is produced per unit of fuel, resulting in greater efficiency. Maximum efficiency is achieved at the engine's rated power, where it operates as intended, maintaining an optimal balance between fuel consumption and power output. Conversely, at lower power levels, the BSFC is higher, meaning that fuel consumption per unit of power is greater, thus reducing the engine efficiency.

The relative efficiency evaluates how the conventional engine is working taking as a reference the optimal efficiency at the nominal operating point as shown in eq. 2.1. At this operating point the relative efficiency is 100% reducing its value if the engine works at lower levels of power.

$$\eta_{\text{rel}} = \frac{\text{BSFC}_{\text{nom}}}{\text{BSFC}} \tag{2.1}$$

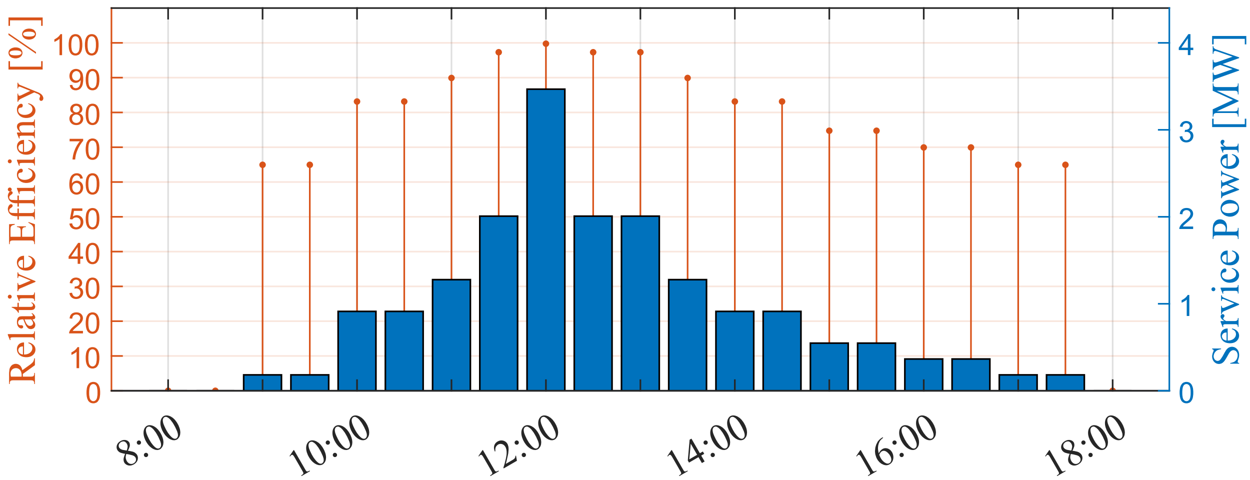


Figure 2.27: Operational profile: one maneuver.

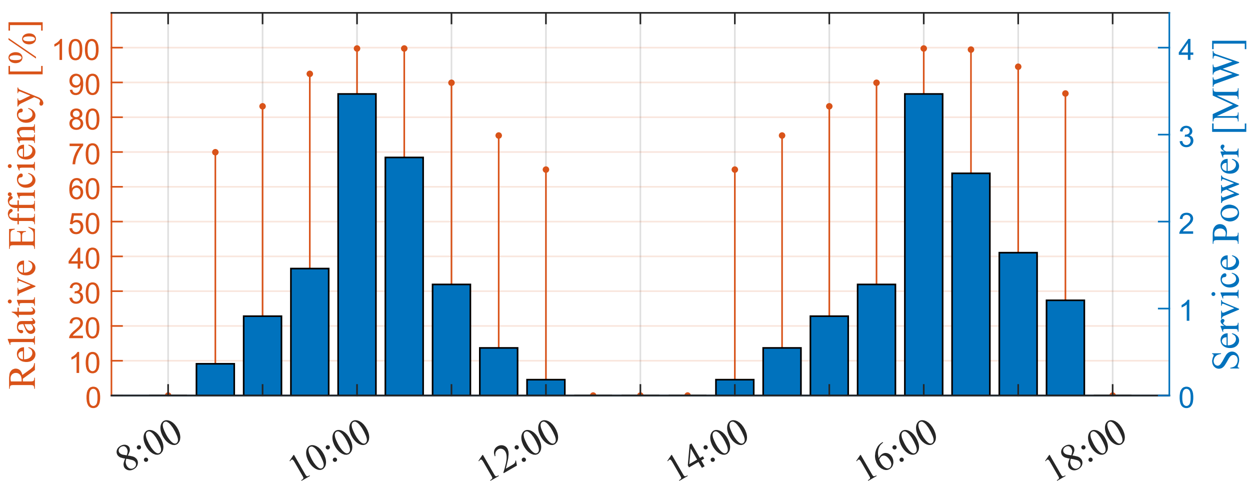


Figure 2.28: Operational profile: two maneuvers.

2.5.2 Generated Emissions and Fuel Consumption

To calculate emissions, the following formula relates brake power (Pb), BSFC, usage time (tu), and a carbon factor (CF):

$$\text{Emissions} = \frac{\text{BSFC} \cdot P_b \cdot t_u \cdot \text{CF}}{10^{-6}} \quad [\text{t CO}_2] \tag{2.2}$$

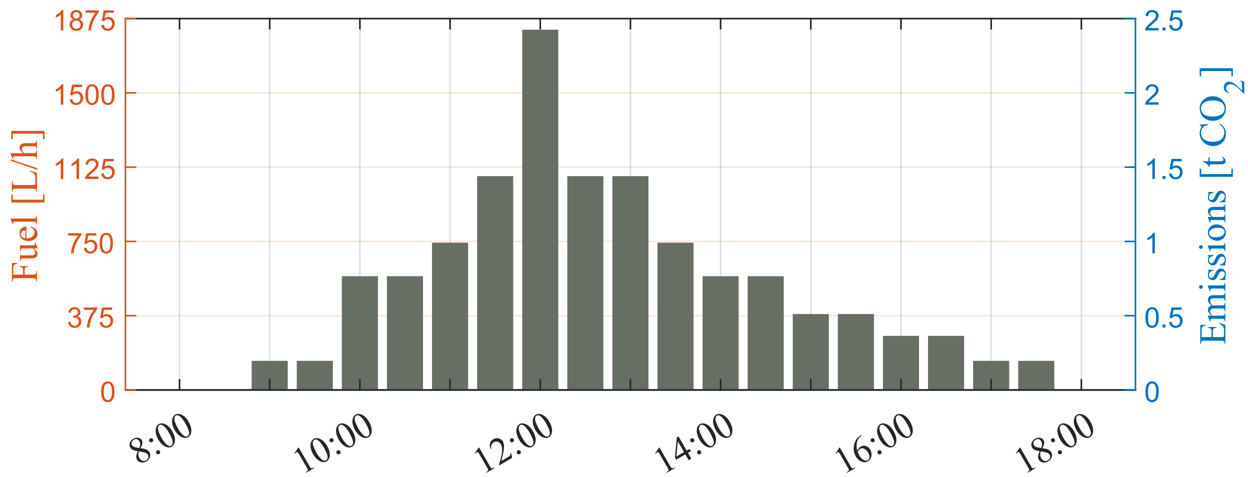


Figure 2.29: Operational profile: one maneuver, fuel consumption and emissions.

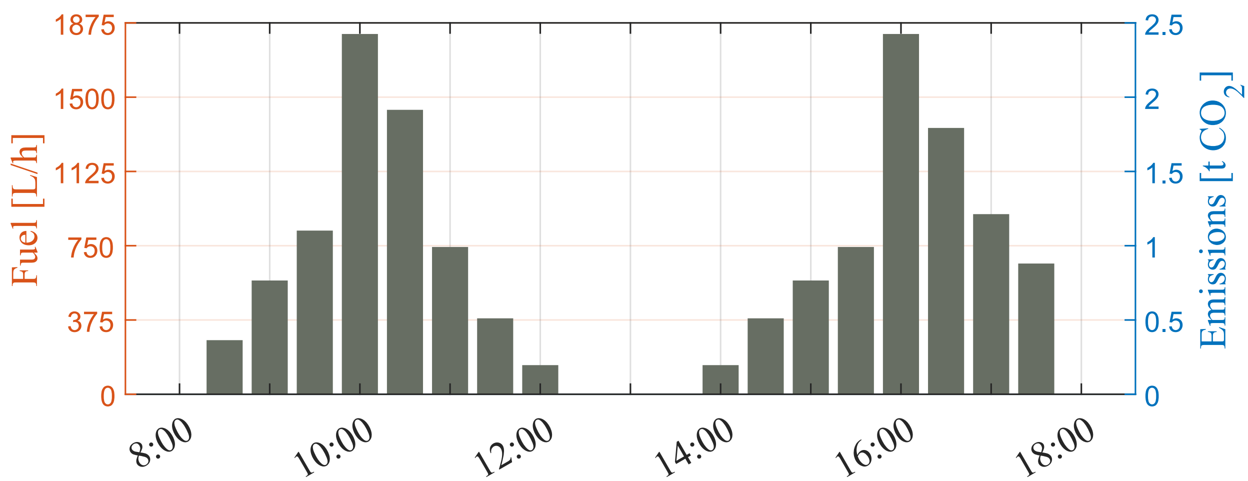


Figure 2.30: Operational profile: two maneuvers, fuel consumption and emissions.

The operational profile of a tugboat operating in a Chilean port typically includes one to two maneuvers per day during a 10-hour work shift, as illustrated in Fig. **??**. Fig. **??**-a) shows the power delivered by the tugboat conventional propulsion system and the relative efficiency of the engines during one service maneuver while b) shows the same information for two maneuvers; c) shows the fuel consumption by the tug and the CO_2 emissions generated by the load of the diesel engines, corresponding to a maneuver, while d) shows the emissions generated by the load of the diesel engines for two service maneuvers.

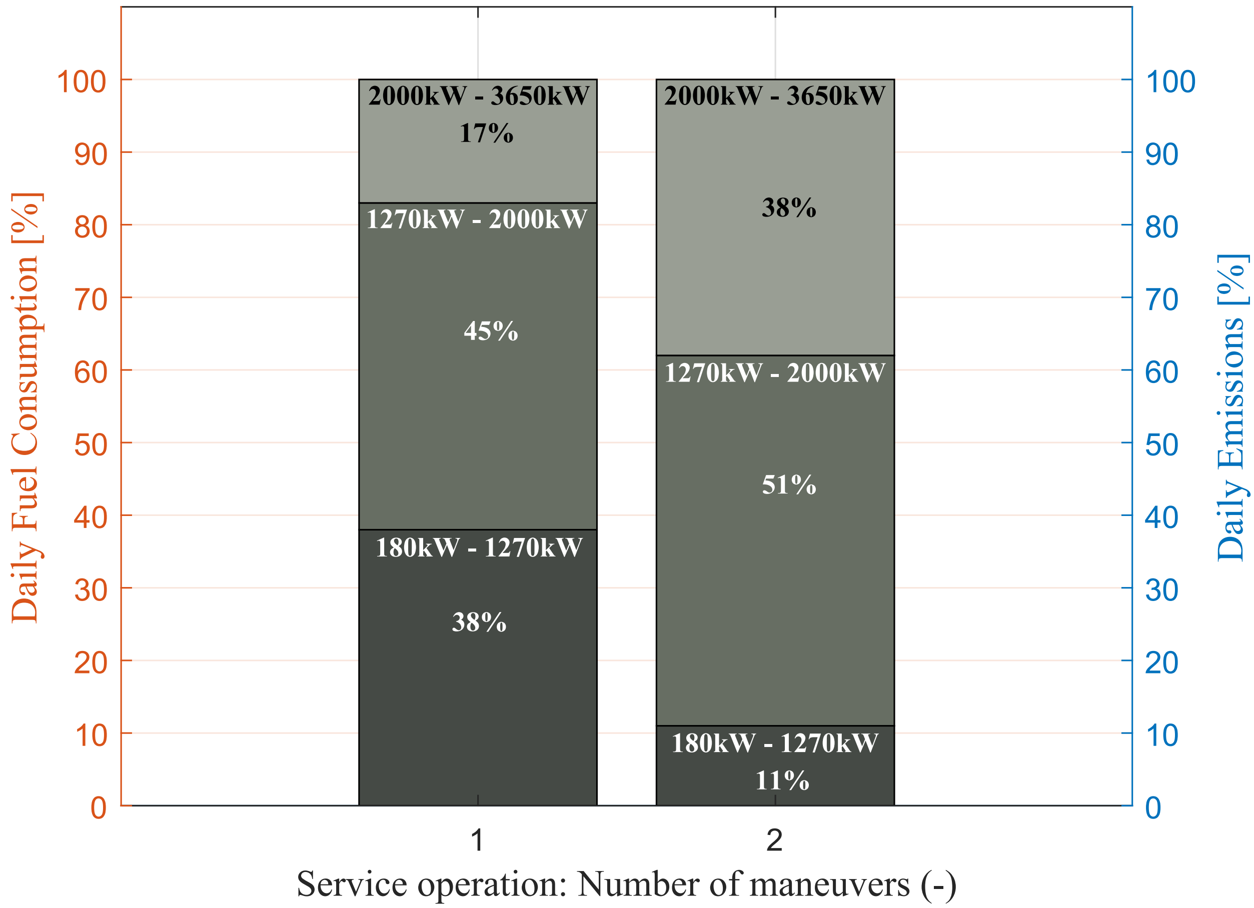


Figure 2.31: Percentage of emissions per day based on power ranges.

It can be observed that as the tugboat operates close to the nominal power point of its diesel engines, its relative efficiency improves. This indicates optimal fuel consumption during those periods. Conversely, at lower operating points, the engine consumes more fuel relative to the energy produced.

Fig. 2.31 shows the accumulated fuel consumption and CO_2 emissions generated for one and two maneuvers. Each service operation is divided into three segments: the lower segments correspond to propulsion system power levels with a relative efficiency below 90%, the middle segments correspond to relative efficiencies ranging from 90% to 98%, and the upper segments represent power levels with the highest relative efficiency, ranging from 98% to 99.8%. At low power levels, fuel consumption can account for up to 38% and 11% of the total daily fuel usage, for one and two maneuvers, respectively. Resulting in high emissions due to the diesel engine operating far from its optimal efficiency point, emphasizing the variability of the load profile of this type of vessel and its impact on fuel consumption and emissions.

2.6 Chapter Summary

Chapter 3

Hybrid Tugboats: Requirements and Performance

This chapter presents the fundamental concepts associated with tugboat hybridization, its most common technological configurations, and the operational benefits associated with the incorporation of battery banks. The objective is to establish the theoretical and technical basis for the design process carried out in subsequent chapters.

3.1 General concepts of marine hybrid systems

3.1.1 Definition of hybrid propulsion system

3.1.2 Classification of hybrid systems

3.1.3 Main Components

3.2 Role of the battery bank in a hybrid system

3.2.1 Main Functions

3.2.2 Advantages of incorporating batteries

3.2.3 Technical limitations

3.3 Impact of hybridization on the operational profile of the tugboat

3.3.1 Fully electric operation at low load

3.3.2 Hybrid operation in medium loads

3.3.3 Optimization of the energy consumption profile

3.3.4 Impact on fuel consumption and emissions

3.4 Case studies and industry evidence

3.4.1 Chapter summary

Chapter 4

Preliminary design of the battery bank for a hybrid propulsion system

This chapter presents the preliminary design process for the battery bank required for the tugboat's hybrid propulsion system. The energy capacity, power requirements, and operating constraints are determined, and then different storage technologies are evaluated from a technical and economic perspective.

4.1 Design parameters and initial assumptions

4.1.1 Operational assumptions

4.1.2 Energy design parameters

4.1.3 Power design parameters

4.1.4 Physical and regulatory constraints

4.2 Battery bank sizing

4.2.1 Calculation of required energy capacity (kWh)

4.2.2 Calculation of required power (kW)

4.3 Battery technologies applicable to tugboats

4.3.1 Selection criteria

4.3.2 Technology analysis

4.4 Economic and life cycle analysis

4.4.1 System CAPEX

4.4.2 System OPEX

4.4.3 ROI / VAN / TIR / Payback

4.5 Comprehensive comparison of alternatives

4.5.1 Selection of the optimal alternative

4.6 Chapter Summary

Chapter 5

Simulation results

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